PHISHING SIMULATION REPORT (LAB ENVIRONMENT)

1. Executive Summary:

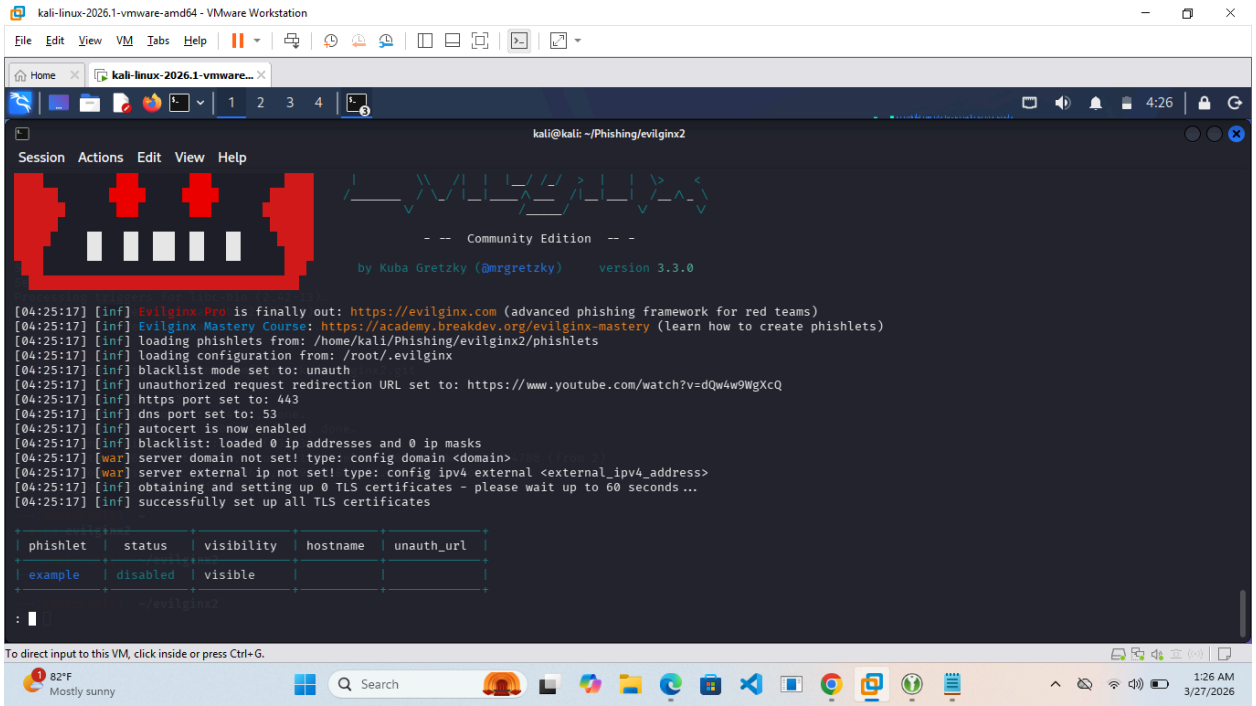
This report documents the execution of a controlled **phishing simulation** conducted within a virtual lab environment using **Gophish** and **Evilginx2**.

The objective was to simulate a real-world phishing attack, evaluate credential harvesting risks, and demonstrate how attackers can capture sensitive user information through social engineering techniques.

The test was performed ethically on a controlled virtual machine (VM), ensuring no real users or systems were impacted.

✅ Key Outcomes:

- Successful deployment of phishing campaign
- Cloned login page hosted via Evilginx2
- Emails sent using Gophish
- Credentials successfully captured and logged
- Risk level identified as **HIGH**



2. Objective:

The primary objectives of this simulation were:

- To set up a phishing campaign using industry tools
- To clone a legitimate login page using Evilginx2
- To deliver phishing emails using Gophish
- To capture and analyze submitted credentials
- To assess the security risk level

3. Tools & Environment:

Tool	Purpose
Gophish	Phishing campaign management and email delivery
Evilginx2	Cloning login pages and capturing credentials
Virtual Machine (VM)	Safe testing environment

4. Methodology:

4.1 Campaign Setup:

A phishing campaign was configured using Gophish. This included:

- Creating an email template
- Importing a target list
- Linking the phishing URL (Evilginx2)
- Launching the campaign

5. Phishing Page Deployment:

A legitimate login page was cloned using Evilginx2.

- The phishing domain was configured
- HTTPS listener was activated
- Victim interaction was tracked

6. Credential Harvesting:

Once targets interacted with the phishing email and entered their credentials, the data was captured and logged.

📊 **Captured Credentials Log:**

Timestamp	IP Address	Username/Password	Risk	Note
2025-08-29 12:00:00	192.168.1.50	testuser/pass123	High	Successful capture

7. Verification (Proof of Exploitation):

To validate the phishing simulation:

- A login attempt was performed on the cloned page
- Credentials were successfully intercepted
- Session/token capture was confirmed

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8. Risk Analysis:

The phishing simulation revealed the following risks:

- Users are highly susceptible to social engineering attacks
- Credentials can be easily harvested through cloned pages
- Lack of user awareness increases vulnerability

🚨 Risk Level: HIGH

Because:

- Credentials were successfully captured
- Attack required minimal technical complexity
- Potential for unauthorized access is significant

9. Key Findings:

- Phishing emails were successfully delivered and interacted with
- Cloned login page appeared legitimate
- Users entered credentials without verification
- Sensitive data was captured in real-time

10. Recommendations:

To mitigate phishing risks:

- Implement **Multi-Factor Authentication (MFA)**
- Conduct **security awareness training** for users
- Deploy **email filtering and anti-phishing solutions**
- Regularly run **phishing simulations**
- Monitor login anomalies and suspicious activity

11. Conclusion:

This phishing simulation successfully demonstrated how attackers can exploit human trust to gain unauthorized access to sensitive information.

The results highlight the importance of cybersecurity awareness, strong authentication mechanisms, and proactive defense strategies.

This lab reinforces the critical need for organizations to continuously test and improve their security posture against phishing attacks.